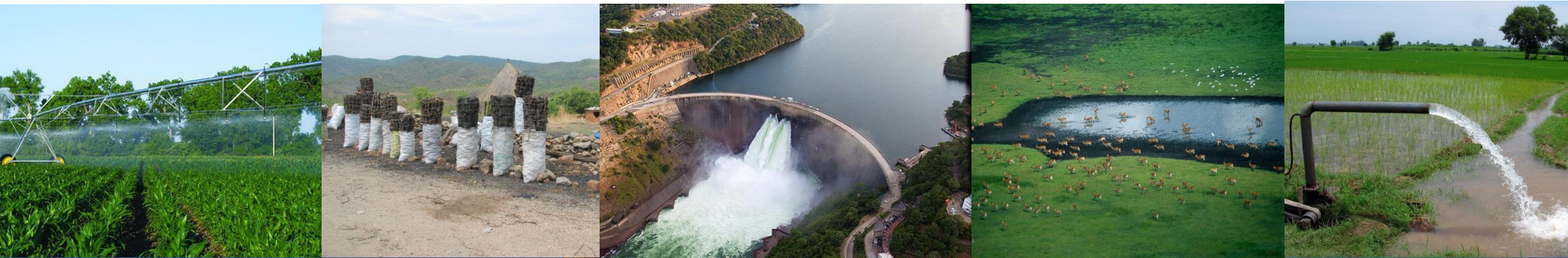


Exercise 2: Hands on CWatM

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Hands on CWatM – Exercise 2

Compare simulated discharge with observed discharge

1. Run CWatM for your basin
2. Look at the output in output
3. Copy output to an Excel sheet
4. Change the settings file and run CWatM again
5. Copy the output again to Excel
6. Explain the difference



Hands on CWatM – Exercise 2

0. download from Sharepoint

SO2/working/CWatM_1min

yourBasin/routing/kinematic/chanleng.nc

(The old file is the same as changrad.nc)



Hands on CWatM

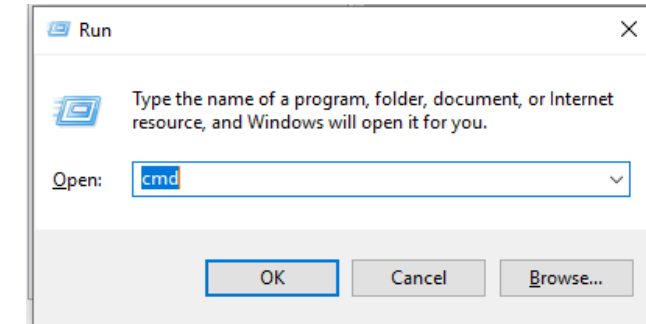
1. Running CWatM

- Go to your basin folder and /Tutorial/Exercise2
- Start: 21_exe_example2.bat

or open a DOS command prompt

- press Windows+R
- type `cmd` + return
- change directory: e.g. `cd c:/CWATM/Exercise2`
(or `cd "c:/directory with white space/CWATM/Exercise2"`)

- Type `python .\CWATM_model\CWatMexe\run_cwatm.py settings_morava_1min_exercise21.ini -l`



```
P:\watmodel\CWATM\Regions\Danube_1min_SP\Morova\Tutorial\Exercise2>python ../../../../CWatM_model_developversion/run_cwatm.py settings_morava_1min_exercise22.ini -l
CWATM - Community Water Model 1.5 Date: 2024/11/27 15:22
International Institute of Applied Systems Analysis (IIASA)
Running under platform: Windows
-----
Create catchment from point and river network
Number of cells in catchment: 4059 = 9080 km2

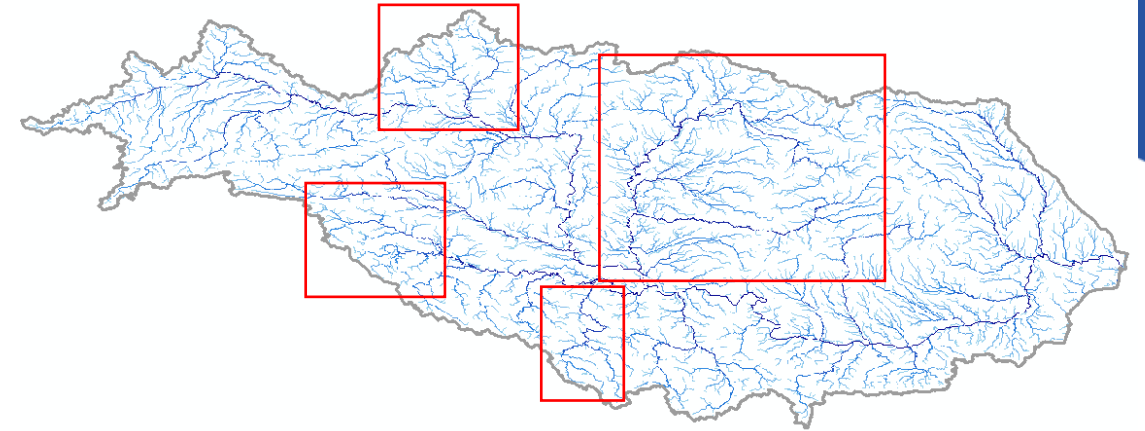
CWATM Simulation Information and Setting
The simulation output as specified in the settings file: P:\watmodel\CWATM\Regions\Danube_1min_SP\Morova\Tutorial\Exercise2/out can be found in settings_morava_1min_exercise22.ini

Step      Date      Discharge
19        19/01/2010    77.56
```

Hands on CWatM

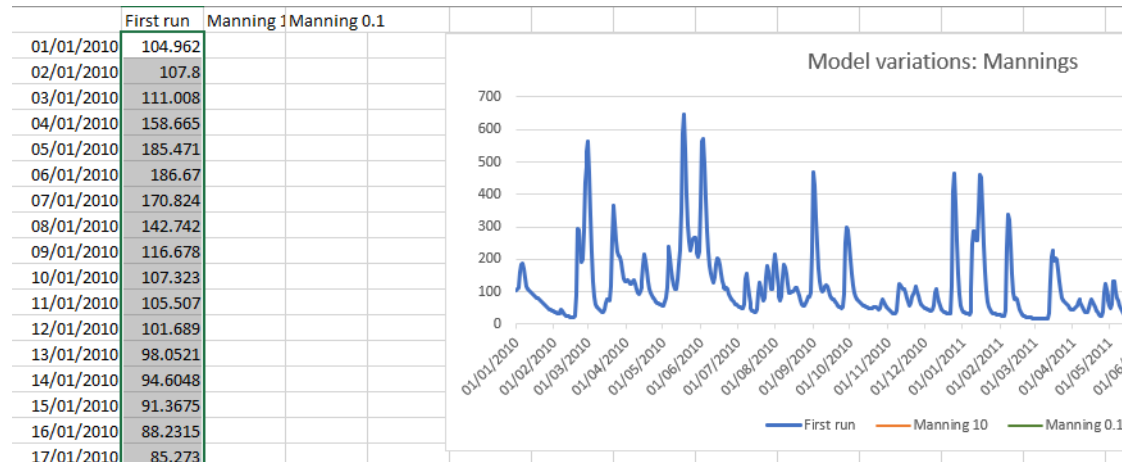
2. Look at the output

- Go into directory ./Tutorial/Exercise2/out
- Open discharge_daily.csv in Excel
- Open ./ Tutorial/Exercise2/exercise2.xlsx
- Copy content of discharge_daily.csv into exercise2.xlsx



Timeseries	settingsfil	Runnung	CWATM: P:\watmodel\CWA		
xloc	17.292				
yloc	48.925				
Date	G1				
01/01/2010	104.962				
02/01/2010	107.8				
03/01/2010	111.008				
04/01/2010	158.665				
05/01/2010	185.471				
06/01/2010	186.67				
07/01/2010	170.824				
08/01/2010	142.742				
09/01/2010	116.678				

out/discharge_daily.csv



exercise2.xlsx

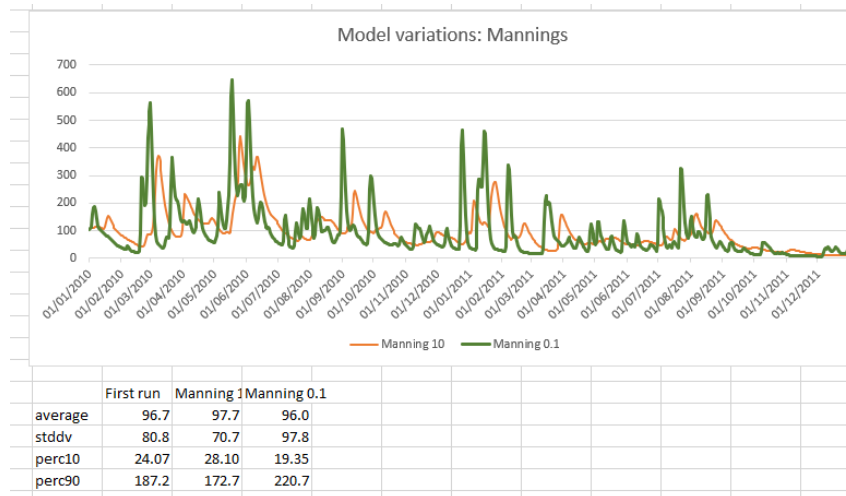
3. Change settings file and run again a settings file

- Change settings file: settings_morava_1min_exercise22.ini
put in: manningsN = 10.0 (routing roughness is increased)
- Run Start: 22_exercise2.bat
- Add results from out/discharge_daily.csv to excel sheet

```
189 [CALIBRATION]
190
191 # These are parameter which are used for calibration
192 # could be any parameter, but for an easier overview, they are collected here
193 # in the calibration template a placeholder (e.g. eta) instead of value
194
195 # reservoir normal storage limit (fraction of total storage, [-]) [0.15 - 0.85] default 0.5
196 # These are parameter which are used for calibration
197 # could be any parameter, but for an easier overview, they are collected here
198 # in the calibration template a placeholder (e.g. 0.11805799193919307eta) instead of value
199
200 SnowFactor = $(FILE_PATHS:PathParameter)/params_snowfactor.map
201 SnowMeltCoef = $(FILE_PATHS:PathParameter)/params_snowMelt.map
202 #IceMeltCoef = $(FILE_PATHS:PathParameter)/params_iceMelt.map
203
204 crop_correct = $(FILE_PATHS:PathParameter)/params_crop.map
205 preferentialFlowConstant = $(FILE_PATHS:PathParameter)/params_pref.map
206 arnoBeta_add = $(FILE_PATHS:PathParameter)/params_arnoB.map
207 ksaf_fact = $(FILE_PATHS:PathParameter)/params_ksafact.map
208
209 factor_interflow = $(FILE_PATHS:PathParameter)/params_interF.map
210 recessionCoeff_factor = $(FILE_PATHS:PathParameter)/params_recres.map
211
212 #manningsN = $(FILE_PATHS:PathParameter)/params_CCM.map
213 manningsN = 10
214
```

3. Change settings file and run again a settings file

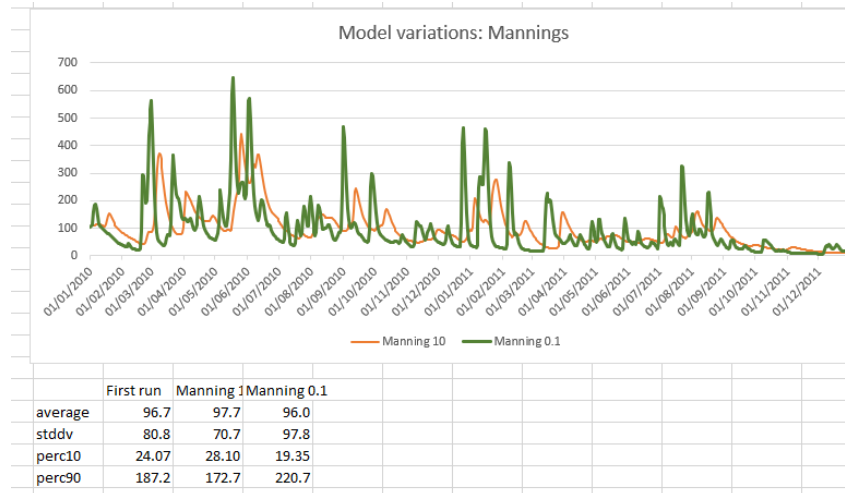
- You can do this step again with Mannings = 0.1



Orange line is 10. Mannings factor run
Green (thicker) line is 0.1 Mannings run

What is the difference?
Why?

Hands on CWatM



Manning's Equation Example

Hydraulic radius (R) = Area / wetted perimeter = $10 \text{ m}^2 / 5 \text{ m} = 2.0 \text{ m}$

Water surface slope = 0.001

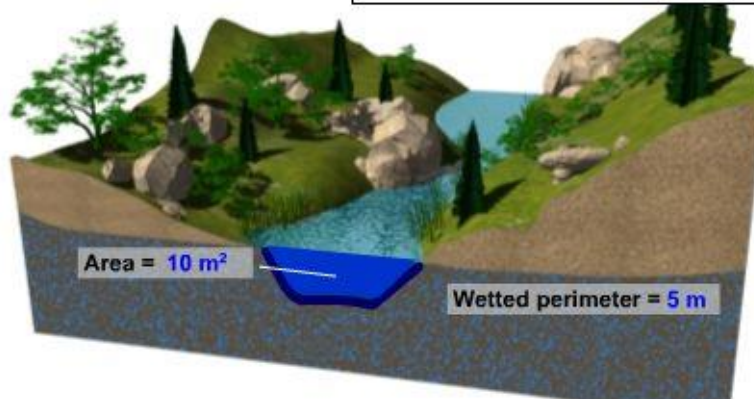
Channel roughness (n) = 0.025

$$V = \frac{R^{2/3} \cdot s^{1/2}}{n}$$

$$V = \frac{2.0^{2/3} \cdot 0.001^{1/2}}{0.025} = 20 \text{ m/s}$$

$$Q = V \cdot A$$

$$Q = 20 \cdot 10 = 200 \text{ cms}$$



Orange line is 10. Mannings factor run
Green (thicker) line is 0.1 Mannings run

Mannings factor is a multiplier to
Mannings' roughness n

Smaller n $\rightarrow$ higher velocity $\rightarrow$ faster $\rightarrow$ higher peak Q

Problems

Most problems come from different file systems, folder structures

We try to set up everything with relative path.

1. Please make sure that your folders have a similar structure like the SharePoint structure

2. The settings file has a part:

[FILE_PATHS]

PathRoot = ../cwatm_data

PathOut = ./output

PathMaps = \$(PathRoot)/cwatm_input30min

PathMeteo = \$(PathRoot)/climate/rhine

../
./

jumps back to the previous folder

uses the folder output in the same folder as the settings file or the directory you are in

3. If this is not working you can use also absolute path (also with white space)

PathRoot = C:/root directory/second.root/cwatm/cwatm_data

4. If you execute cwatm you can also use absolute path

instead

../CWATM_model/CWatMexe/cwatm.exe settings_rhine30min.ini -l

"C:/root directory/second.root/cwatm/CWATM_model/CWatMexe/cwatm.exe" settings_rhine30min.ini -l (mind the " if there are white spaces)

5. Your textfile is not in **UTF-8** (Unicode Transformation Format - 8-bit). Can happen if you use the Windows standard texteditor (wordpad, notepad). Please use an advanced texteditor (e.g. Notepad++)

6. Some other errors we address in:

<https://cwatm.iiasa.ac.at/tutorial.html#test-the-python-model-version>